



What is a Wordlist?

A **wordlist** is a plain text file that contains a list of potential passwords—one per line—used in **brute-force** or **dictionary attacks** to crack passwords or hashes.

Common Use Cases:

- Cracking passwords using tools like **John the Ripper**, **Hydra**, **Hashcat**, etc.
- Penetration testing.
- Password auditing.



What is Crunch?

Crunch is a powerful command-line tool in Kali Linux and other distros used to **generate custom wordlists** with specific patterns, character sets, lengths, etc.



Crunch Basic Syntax

```
crunch <min_length> <max_length> [options]
```

Example:

```
crunch 4 6
```

This generates all possible combinations from length 4 to 6 using default charset (lowercase letters).

◆ 1. Lowercase letters only (length 4–6)

```
crunch 4 6 abcdefghijklmnopqrstuvwxyz -o wordlist.txt
```



Breakdown:

- 4 6: Passwords will be **minimum 4** and **maximum 6** characters long

- abcdefghijklmnopqrstuvwxyz: Only lowercase letters used
- -o wordlist.txt: Output will be saved to a file named wordlist.txt

Example Output:

```
aaaa
aaab
aaac
...
zzzzyz
zzzzyz
zzzzzz
```

◆ **2. Include uppercase, lowercase, numbers, and symbols**

```
crunch 6 8
abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890
!@# -o strong.txt
```

Breakdown:

- 6 8: Passwords of length **6 to 8**
- Character set: Lowercase + Uppercase + Digits + Symbols (!@#)
- -o strong.txt: Saves to strong.txt

Use Case:

Useful when you're targeting systems with **strong password policy** (complex passwords).

◆ **3. Use built-in charset (mix of alpha and numeric)**

```
crunch 6 6 -f /usr/share/crunch/charset.lst mixalpha-numeric -o mix.txt
```

Breakdown:

- 6 6: Fixed length passwords of 6
- -f: Use predefined character sets from file /usr/share/crunch/charset.lst
- mixalpha-numeric: Uses lowercase, uppercase, and numbers

- -o mix.txt: Output file

Tip:

You can explore /usr/share/crunch/charset.lst for more sets like:

- alpha
- numeric
- mixalpha-numeric-all-space
- symbols

◆ **4. Pattern-based wordlist (specific format)**

```
crunch 8 8 -t @@11@@11 -o pattern.txt
```

Breakdown:

- 8 8: Passwords of exactly 8 characters
- -t @@11@@11: Pattern format
- @ = lowercase, , = uppercase, % = number, ^ = symbol
- -o pattern.txt: Saves to pattern.txt

What @@11@@11 means:

- @ = lowercase letter
- 1 = number (digit)

So @@11@@11 generates passwords like:

ab12cd34

xy99az01

...

Crunch Pattern Symbols Quick Reference

Symbol Meaning

@	Lowercase letter
,	Uppercase letter
%	Numbers (0–9)
^	Symbols (!@#\$...)



Extra Tips:

- ◆ **Save Huge Wordlists to Compressed File (to save space):**

```
crunch 8 8 abc123 | gzip > wordlist.gz
```

- ◆ **Redirect output to stdout for piping into other tools:**

```
crunch 6 6 abc123 | john --stdin hashfile
```

- ◆ **Avoid writing to file (for very large combinations):**

Just use stdout and pipe directly to tool like Hydra:

```
crunch 4 4 1234567890 | hydra -l admin -P - 192.168.1.10 ssh
```